Infrastructure
Les 2a

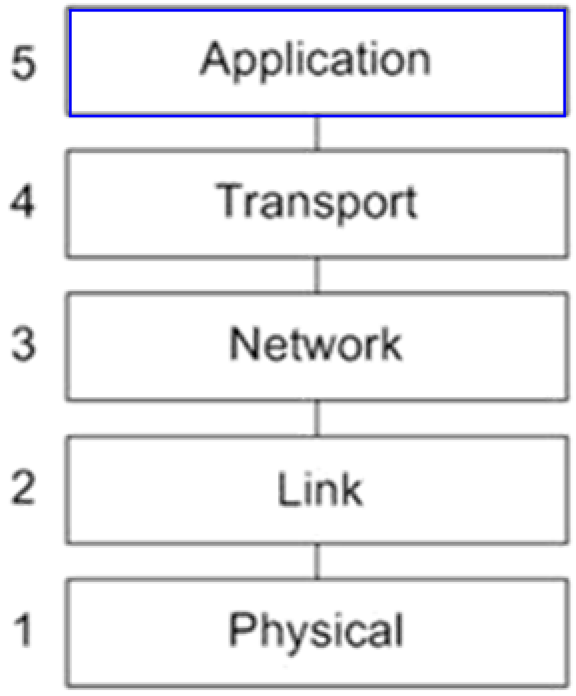
DNS & Transport Protocol

HBO-ICT | Hogeschool van Amsterdam

Creating Tomorrow

Agenda

- Application Layer
- Application protocols
- Sockets en port nummers
- Application protocol voorbeelden
- Netwerkarchitecturen



TCP/IP Model

DNS

- DNS services
- DNS servers
- DNS records

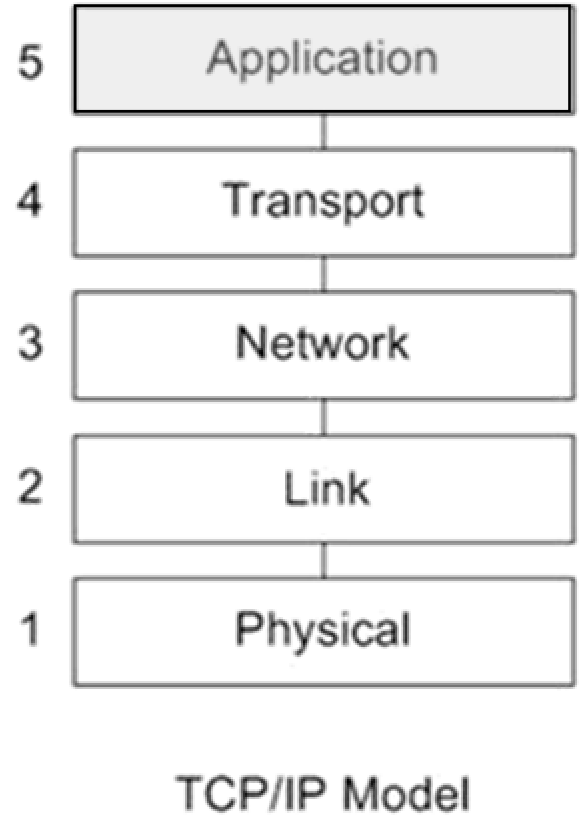


Figure 1

DNS: domain name system

•*people:* many identifiers:

- SSN, name, passport #

•*Internet hosts, routers:*

- IP address (32 bit) - used for addressing datagrams
- “name”, e.g., www.yahoo.com - used by humans

•*Q:* how to map between IP address and name, and vice versa ?

•*Domain Name System:*

•*distributed database* implemented in hierarchy of many *name servers*

•*application-layer protocol:* hosts, name servers communicate to *resolve* names (address/name translation)

- note: core Internet function, implemented as application-layer protocol
- complexity at network’s “edge”

DNS: services, structure

- *DNS services*
- hostname to IP address translation
- host aliasing
  - canonical, alias names
- mail server aliasing
- load distribution
  - replicated Web servers: many IP addresses correspond to one name

- *why not centralize DNS?*
- single point of failure
- traffic volume
- distant centralized database
- maintenance

DNS: a distributed, hierarchical database

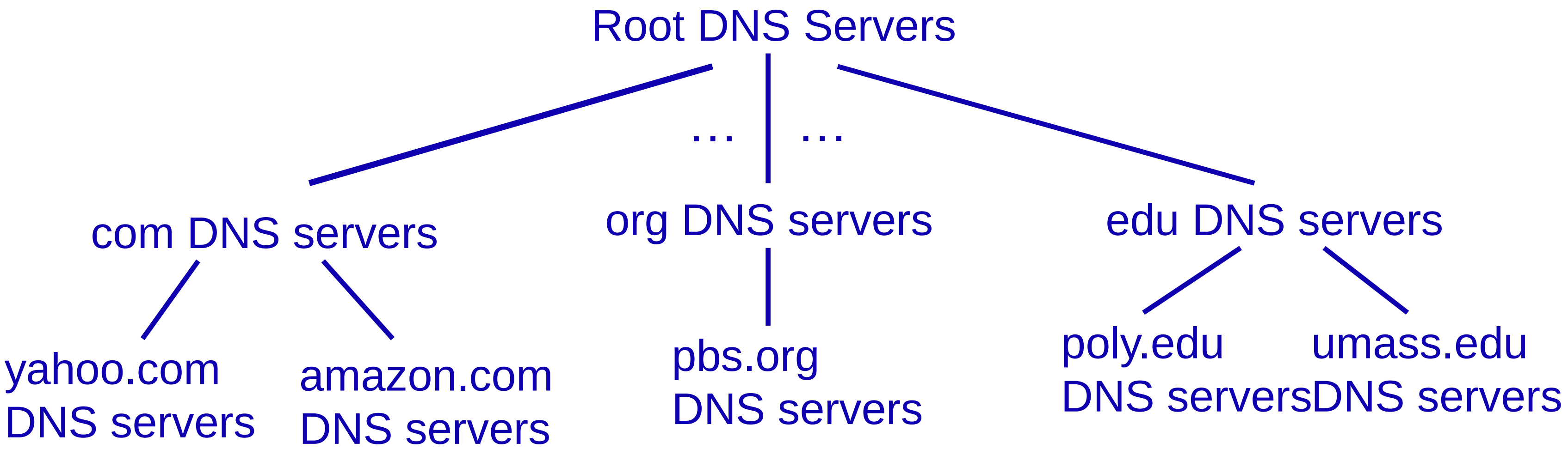
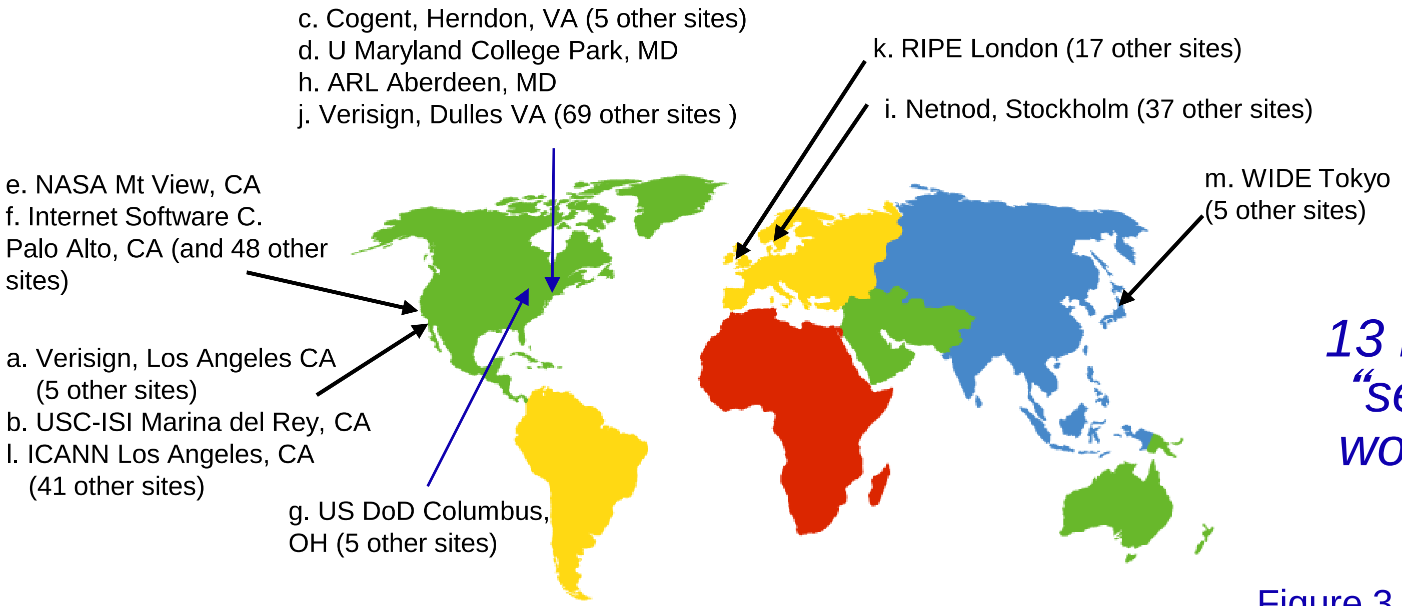


Figure 2

- *client wants IP for www.amazon.com:*
- client queries root server to find .com DNS server
- client queries .com DNS server to get amazon.com DNS server
- client queries amazon.com DNS server to get IP address for www.amazon.com

DNS: root name servers

- contacted by local name server that can not resolve name
- root name server:
  - contacts authoritative name server if name mapping not known
  - gets mapping
  - returns mapping to local name server



13 root name "servers" worldwide

Figure 3

TLD, authoritative servers

- *top-level domain (TLD) servers:*
  - responsible for com, org, net, edu, aero, jobs, museums, and all top-level country domains, e.g.: uk, fr, ca, jp

- *authoritative DNS servers:*
  - organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts
  - can be maintained by organization or service provider

Local DNS name server

- does not strictly belong to hierarchy
- each ISP (residential ISP, company, university) has one
  - also called "default name server"
- when host makes DNS query, query is sent to its local DNS server
  - has local cache of recent name-to-address translation pairs (but may be out of date!)
  - acts as proxy, forwards query into hierarchy

DNS name resolution example

•host at cis.poly.edu wants IP address for gaia.cs.umass.edu

iterated query:

- contacted server replies with name of server to contact
- "I don't know this name, but ask this server"

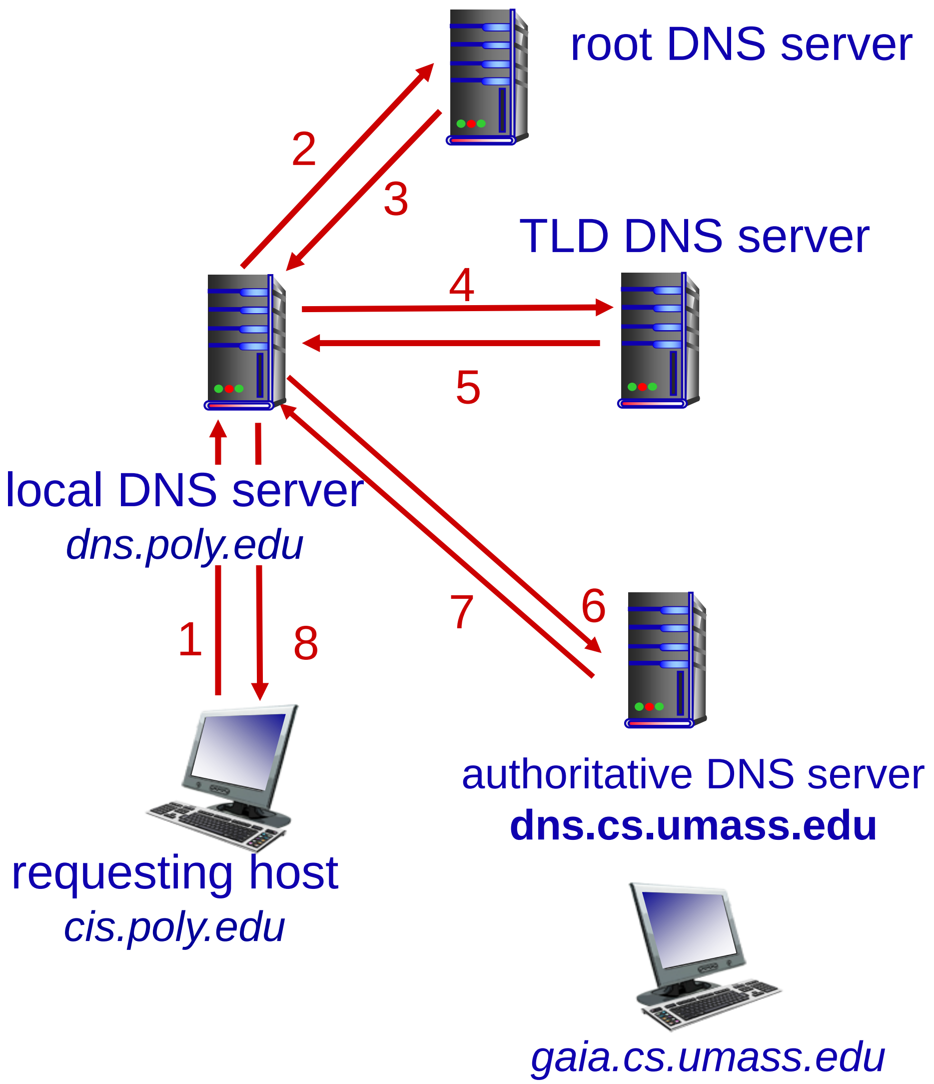


Figure 4

DNS records

DNS servers store different types of resource records:

- A-record: hostname and IP-address for this host
- CNAME-record: alternative names (aliases) for a hostname
- NS-record: name of an authoritative nameserver for a domain
- MX-record: name or alternative names for a mailserver that handles mail for a domain

DNS: caching, updating records

- once (any) name server learns mapping, it *caches* mapping
  - cache entries timeout (disappear) after some time (TTL)
  - TLD servers typically cached in local name servers
    - thus root name servers not often visited
- cached entries may be *out-of-date* (best effort name-to-address translation!)
  - if name host changes IP address, may not be known Internet-wide until all TTLs expire

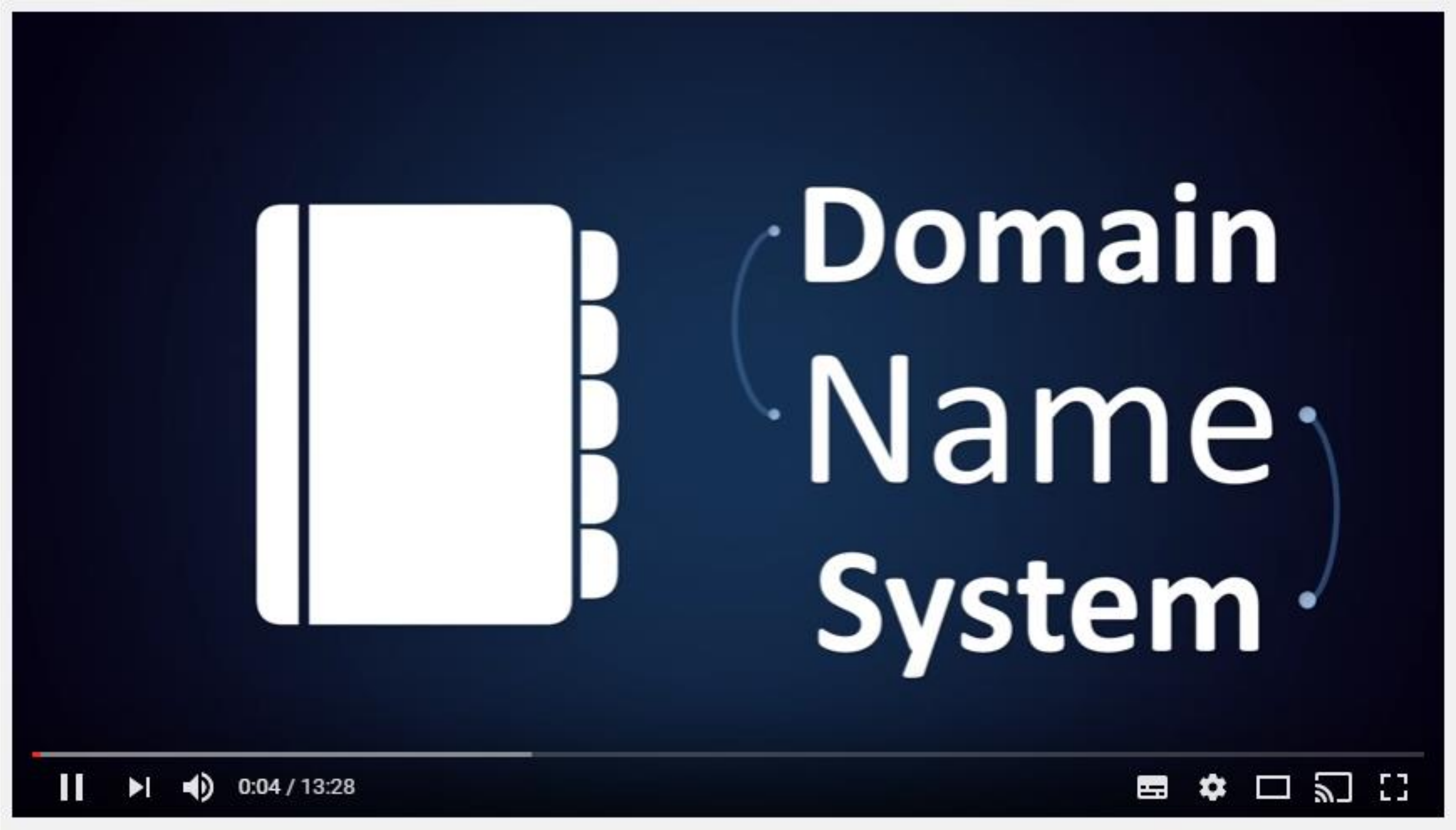
DNS spoofing & DNSsec

Figure 15

- \- DNS is easy to spoof
  - capture the request
- \- reply with your own IP address
- \- catch the HTTP/Email/.. etc request and have your way with it..

- \- in DNSsec the DNS reply is accompanied by a digital certificate
- \- the certificate is signed by a Certificate Authority
- \- therefore much harder to spoof
- (in security we never say ’impossible')

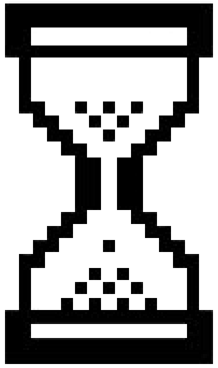
DNS



[Link to movie]

Figure 5

Break



Transport layer

- Transport layer services
- UDP
- TCP
- The TCP connection
  - flow control
  - congestion control
  - pipelining
  - reliable data transfer

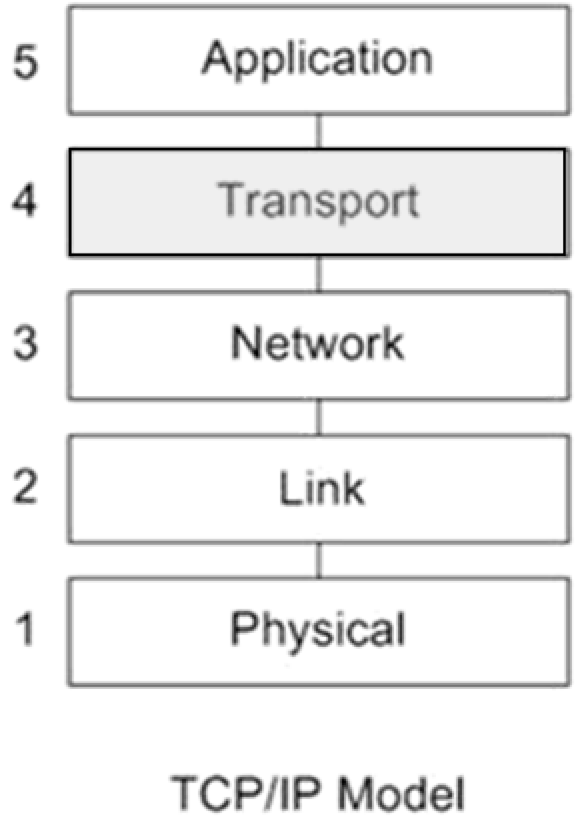


Figure 6

Transport services and protocols

- provide *logical communication* between app processes running on different hosts
- transport protocols run in end systems
  - sending side: breaks app messages into *segments*, passes to network layer
  - receiving side: reassembles segments into messages, passes to app layer
- more than one transport protocol available to apps
  - Internet: TCP and UDP

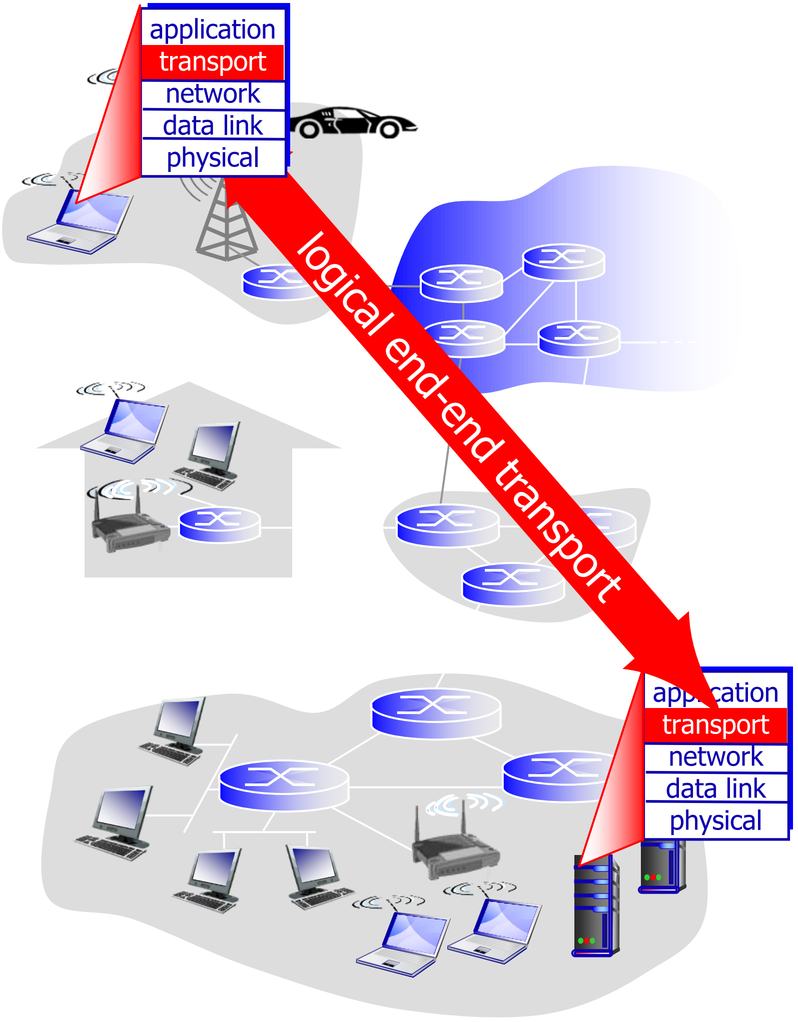


Figure 7

Transport vs. network layer

- *network layer:* logical communication between hosts

- *transport layer:* logical communication between processes
  - relies on, enhances, network layer services

Analogy:

- *You send a letter to your teacher, addressed to Theo Thijssenhuis, Amsterdam*
- *There are many teachers in this building!*
- *So each teacher has their own pigeon hole*

- *host = building*
- *processes = teachers*
- *mailbox = socket with port number*
- *network-layer protocol = postal service*
- *transport protocol = faculty staff that sorts the letters*

Internet transport-layer protocols

- unreliable, unordered delivery: UDP
  - no-frills extension of "best-effort" IP
- reliable, in-order delivery (TCP)
  - congestion control
  - flow control
  - connection setup
- services not available:
  - delay guarantees
  - bandwidth guarantees

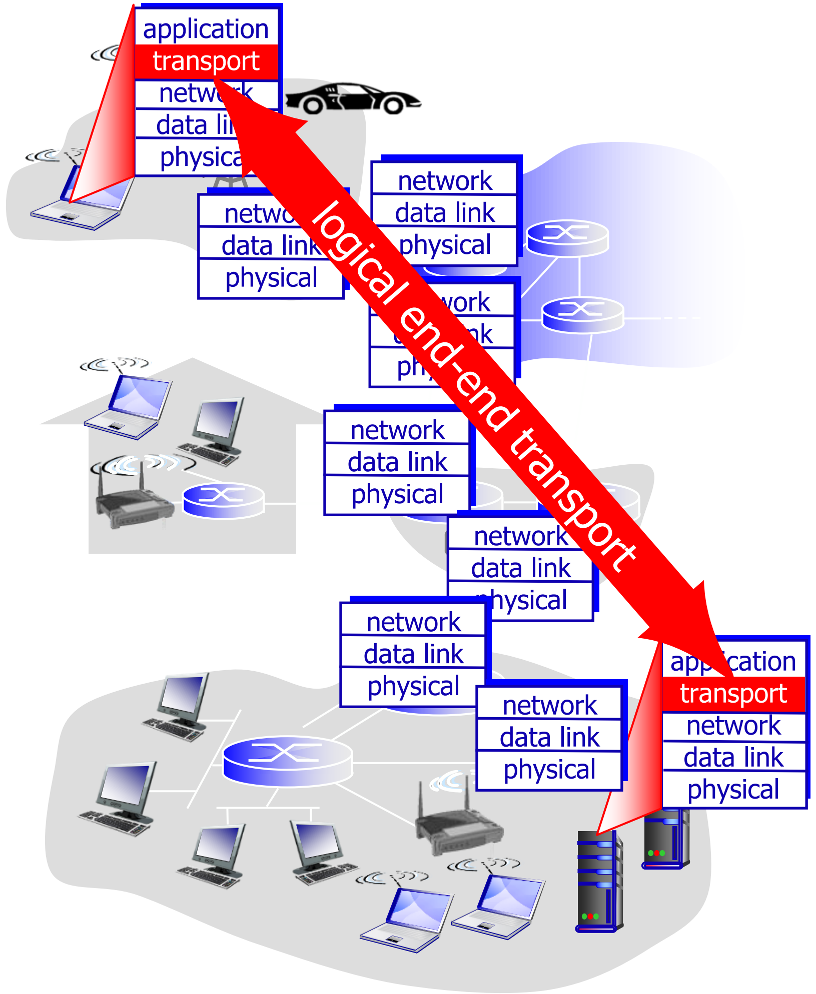


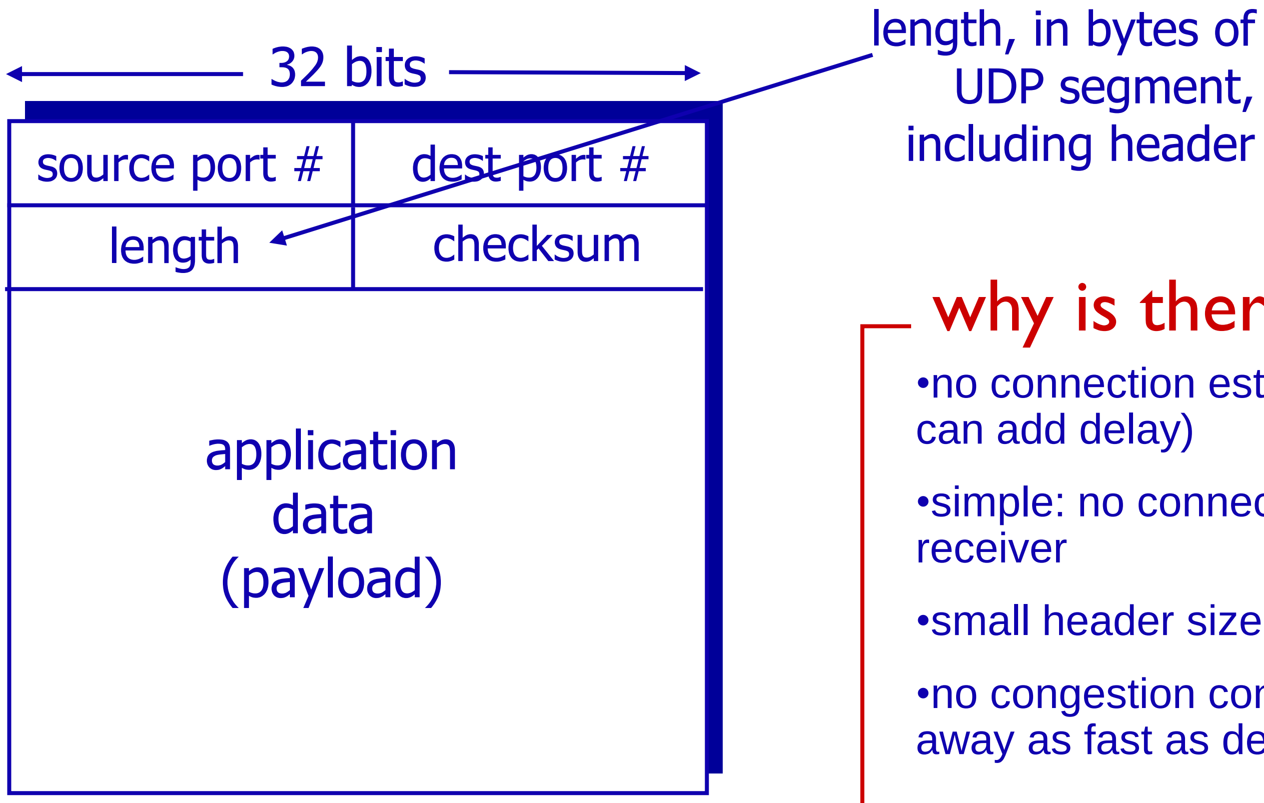
Figure 8

UDP: User Datagram Protocol [RFC 768]

- "no frills," "bare bones" Internet transport protocol
- "best effort" service, UDP segments may be:
  - lost
  - delivered out-of-order to app
- *connectionless:*
  - no handshaking between UDP sender, receiver
  - each UDP segment handled independently of others

- UDP use:
  - streaming multimedia apps (loss tolerant, rate sensitive)
  - DNS
  - SNMP

UDP: segment header



UDP segment format

Figure 9

why is there a UDP?

- no connection establishment (which can add delay)
- simple: no connection state at sender, receiver
- small header size
- no congestion control: UDP can blast away as fast as desired

TCP: Overview

RFCs: 793,1122,1323, 2018, 2581

- point-to-point:
  - one sender, one receiver
- connection-oriented:
  - handshaking (exchange of control msgs) inits sender, receiver state before data exchange
- reliable, in-order *byte stream:*
  - no "message boundaries"
- flow control:
  - sender will not overwhelm receiver
- congestion control:
  - sender will not overwhelm network
- pipelined:
  - TCP congestion and flow control set window size
- full duplex data:
  - bi-directional data flow in same connection

TCP segment structure

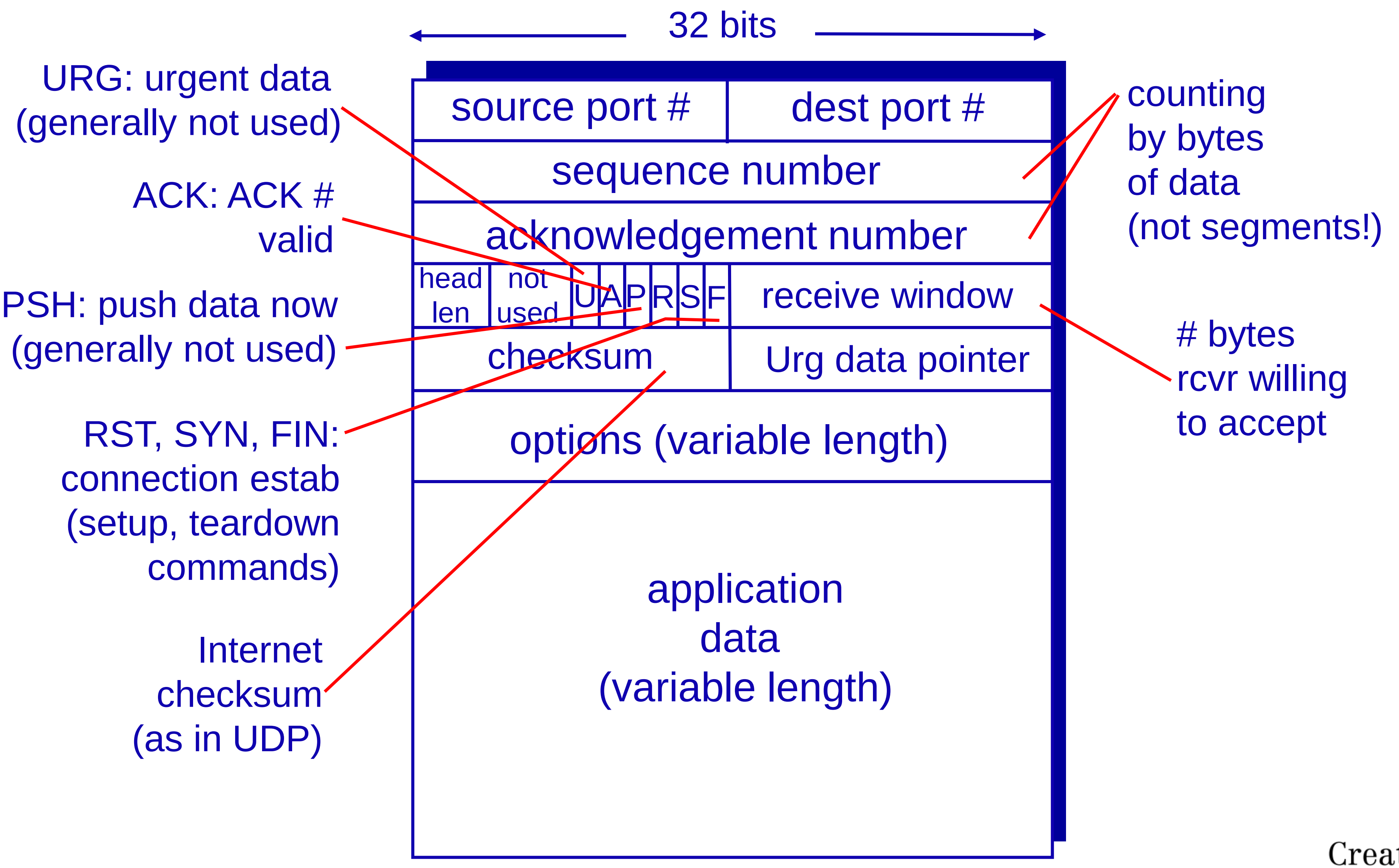


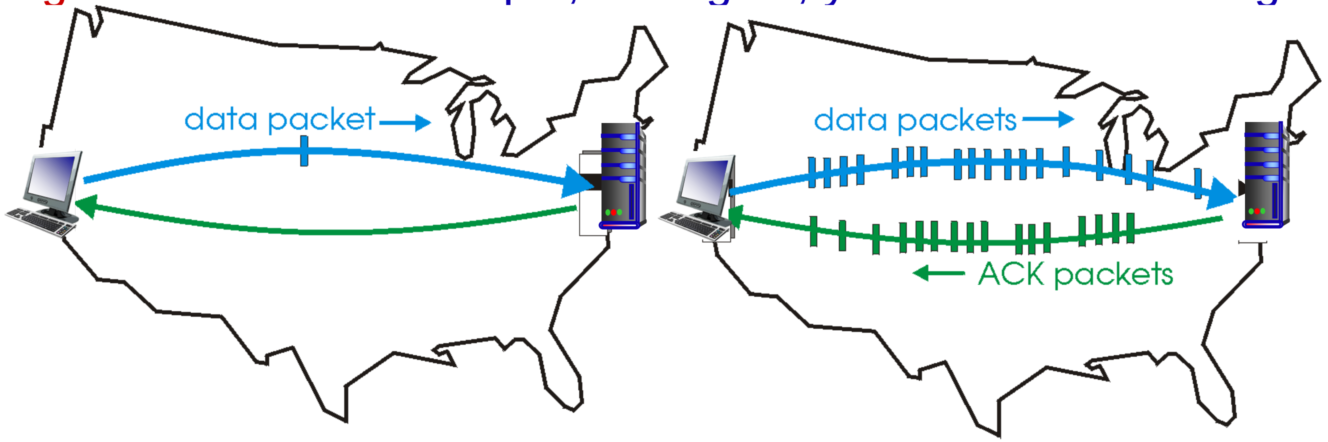
Figure 10

Reliable data transfer

- How does TCP accomplish reliable data transfer?
  - For each packet of data sent, TCP expects to receive a confirmation from the receiver
  - Otherwise, the data is sent again

- This can be slow if you wait for each packet to be confirmed!

- pipelining: sender allows multiple, “in-flight”, yet-to-be-acknowledged pkts



(a) a stop-and-wait protocol in operation (b) a pipelined protocol in operation

Figure 11

TCP Seq & ACK

TCP keeps track of the number of bytes that are transmitted and confirmed

- The SEQUENCE number indicates the number of bytes transmitted
- The ACKNOWLEDGEMENT number indicates the number of bytes received correctly, and that the receiver is ready for the next byte

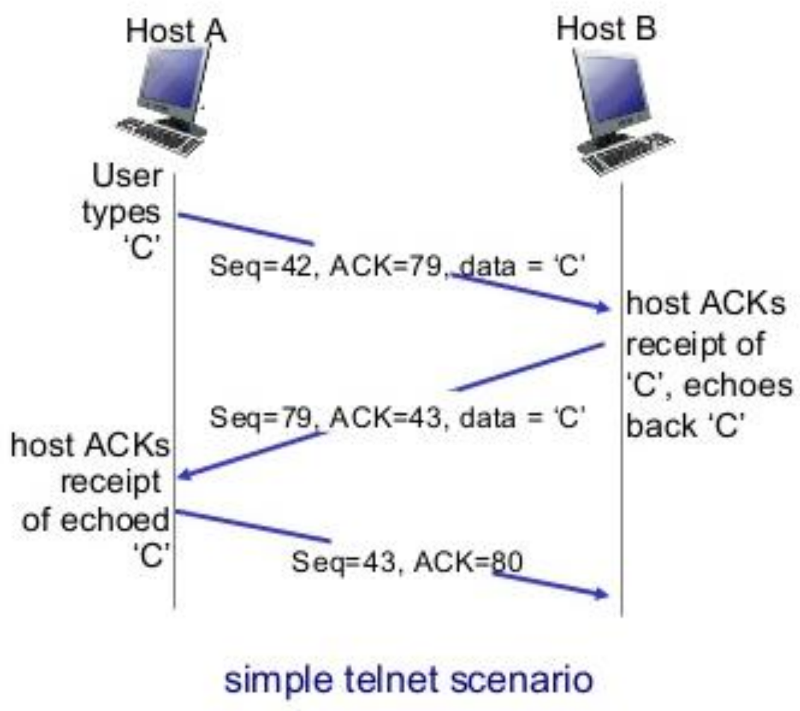


Figure 12

TCP loss scenario

If the correct reception is not confirmed in time, data will be retransmitted

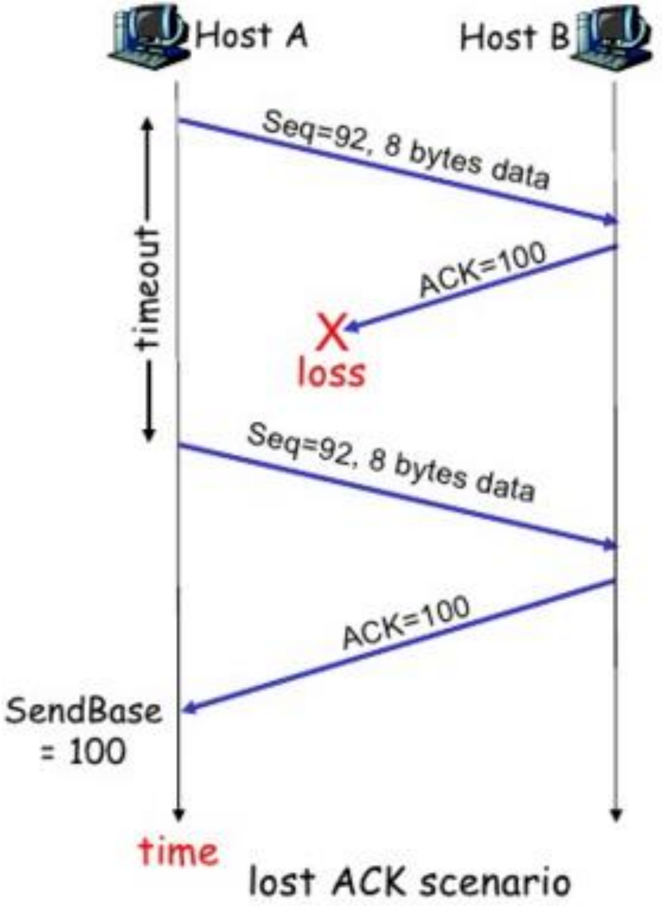
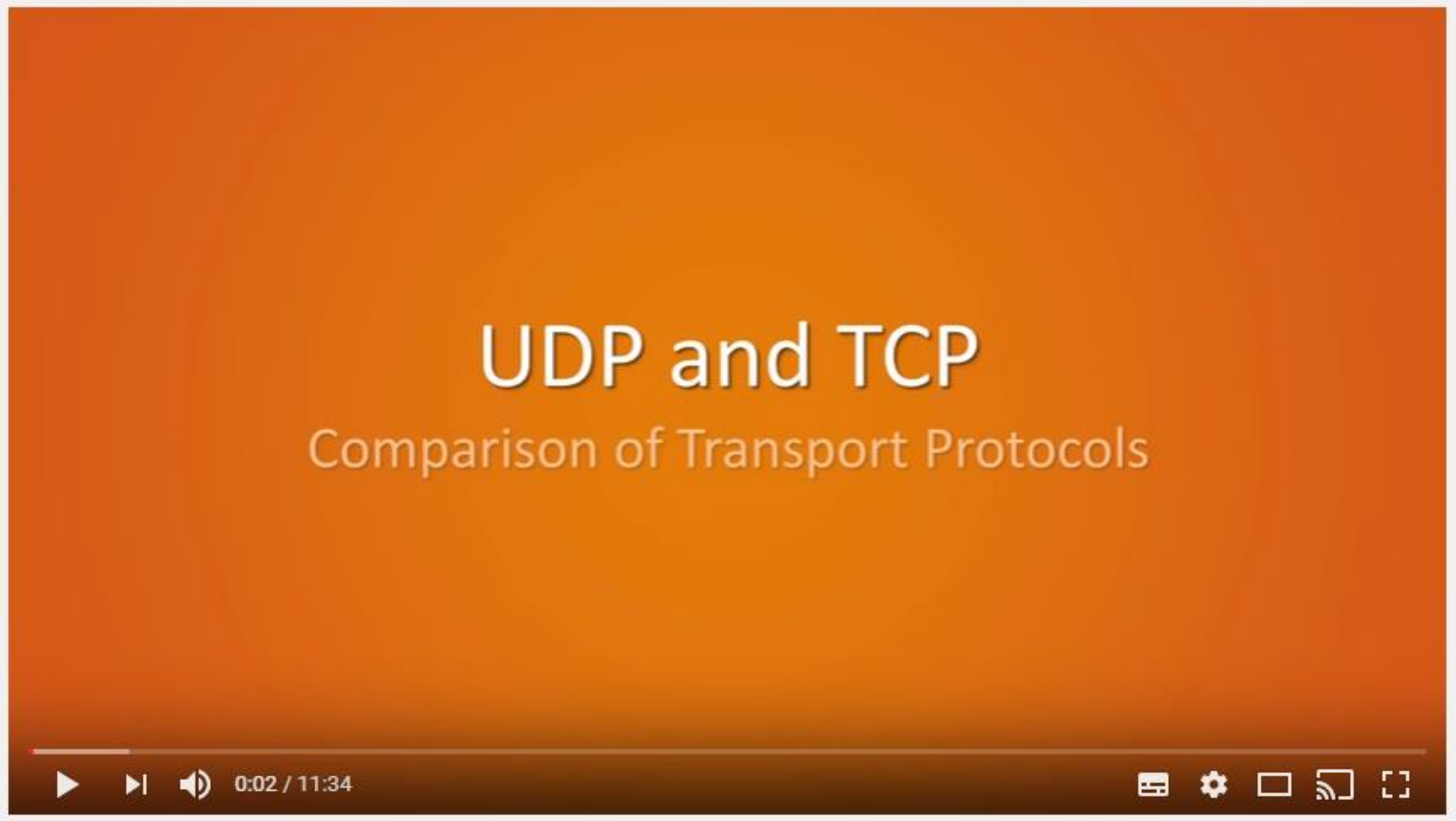


Figure 13

UDP & TCP compared



[Link to movie]

Figure 14

Sources

- Figure 1,2,3,4,6,7,8,9,10,11,12,13: J.F. Kurose and K.W. Ross, Computer Networking, A Top-Down Approach, Pearson Education, 2013

- Figure 5: PieterExplainsTech, 28-8-2014, Inside the Domain Name System - YouTube Video, retrieved on 7-3-2017 from https://www.youtube.com/watch?v=GlZC4Jwf3xQ

- Figure 14: PieterExplainsTech, 24-7-2013, UDP and TCP: Comparison of Transport Protocols - YouTube Video, retrieved on 7-3-2017 from https://www.youtube.com/watch?v=Vdc8TCESIg8

- Figure 15, jaydeep_, February 1, 2018, A padlock superimposed over a blue circuit board pattern - online image, retrieved on 9-9-2018 from https://en.wikipedia.org/wiki/Computer_security#/media/File:Cybersecurity.png